

Recall from last class:

- An n by n matrix A is **invertible** if there is an n by n matrix C so that $AC = I_n$ and $CA = I_n$. We denote the inverse of A by A^{-1} .
- **Theorem:** Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.
- We define the **determinant** of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, denoted by $\det(A)$, by $\det(A) = ad - bc$.

Let S be a non-empty set.

Definition 1. A set S is **finite** if there exists a bijection (one-to-one correspondence) $f : S \rightarrow \{1, 2, \dots, n\}$ for some natural number n . The set S is **infinite** if no such bijection exists for any n .

Remark: We also call the empty set finite.

Definition 2. The **size** or **cardinality** of a finite set S is the unique n such that there exists a bijection from S to $\{1, 2, \dots, n\}$. We denote the cardinality of S by $|S|$.

Remark: We also define the cardinality of the empty set to be 0.

1. We've seen the formula

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Find a formula for $|A \cup B \cup C|$.

2. Can you conjecture a more general formula for $|A_1 \cup A_2 \cup \dots \cup A_k|$?

Definition 3. *Sets A and B have the same cardinality if there is a bijection from A to B .*

3. How do the sets $\mathbb{N}$ and $2\mathbb{N} = \{2n : n \in \mathbb{N}\}$ compare?

4. How do the sets $\mathbb{N}$ and $\mathbb{N}^2$ compare?

Definition 4. *An infinite set S is called **countably infinite** if there is a bijection from S to $\mathbb{N}$; otherwise S is **uncountably infinite** or **uncountable**.*

5. Is the set $\mathbb{Z}$ countably infinite?

6. Compare the following sets.

(a) $A = (0, \infty)$ and $B = \mathbb{R}$.

(b) $A = (0, 1)$ and $B = \mathbb{R}$.

7. Compare $\mathbb{R}$ and $\mathbb{N}$.

Definition 5. Define the **power set** of a set A as the set of all subsets of A ; namely

$$P(A) = \{X : X \subseteq A\}.$$

8. Let $A = \{1, 2\}$. Find the power set of A .

9. What is $|P(\{1, \dots, n\})|$?

Theorem: There is no bijection from A to $P(A)$.

10. Prove the Theorem.